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Front cover: Henlein medal, 1905. He is shown as a conqueror, the portable watch in his left hand being a victory over the 'old' immovable weightdriven clocks, represented by the weight with a chain under his right foot. Photo: Germanisches Nationalmuseum, Nuremberg.

Early watches—The argument over priority in Italy and Germany

Günther Oestmann*

Peter Henlein – Nuremberg – the world's first pocket watch: these three notions seem to be inextricably linked, but were (and are) heavily contested. These disputes are described briefly, with special consideration of the life and work of Enrico Morpurgo, professor of Italian in Amsterdam (1894–1972).

The invention of the first pocket watch by Peter Henlein of Nuremberg (Fig. 1) is taken by the general public as an historical fact. In Germany many streets and squares have been named after him, and a number of monuments survive (Fig. 2). But the history of the 'invention' of the pocket watch is a conglomerate of undisputed original sources and more or less audacious conjectures laced with a good input of local patriotism, and the zealous production of myths.

Already in 1730 the astronomer Johann Gabriel Doppelmayr (1677–1750) took it for granted that the invention of the pocket watch took place in Nuremberg, quoting the famous reference to Henlein in the *Cosmography* of Pomponius Mela edited by Johann Cochläus in 1512:

Each day more subtle things are invented. Thus Peter Hele, still quite a young man, makes works which even very learned mathematicians [cosmographers] admire, for from a little bit of iron he makes clocks composed of many wheels which, however wound, will run without weights and show the time for 40 hours, and which may be carried on the person or in a purse.¹



Fig. 1. Henlein medal struck by Ludwig Christoph Lauer in Nuremberg, 1905 (Bronze, diameter 50,4 mm; Schramberg, Junghans GmbH, inv. nr. 4584). In the iconography of the period, he is shown as a conqueror, the portable watch in his left hand being a victory over the 'old' immovable weight-driven clocks, represented by the weight with a chain under his right foot.

There are no objects extant which can be ascribed to Henlein with certainty. Cochlaeus did not specify the shape of Henlein's watches,

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1. 'Inveniunt in dies subtiliora, etenim Petrus Hele juvenis adhuc admodum opera fecit, quae vel doctissimi admirantur Mathematici, nam ex ferro parva fabricat horologia plurimis digesta rotulis, quae quocunque vertantur, absque ullo pondere & monstrant et pulsan XL. horas, etiamsi in sinu marsupiove contineantur'; J. G. Doppelmayr, *Historische Nachricht von den Nürnbergischen Mathematicis und Künstlern*. Nuremberg, P. C. Monath, 1730, p. 286.

2. '15 fl. H. Henlein für 1 vergulden pyn Apfel für all Ding mit einem Oaiologium'; A. Gümbel, *Peter Henlein der Erfinder der Taschenuhren*, Halle, Zentralverband der Deutschen Uhrmacher, 1924, 27, n. 29.

but a record of payment of 1524 for a watch housed inside a musk apple (*Bisamapfel*) is extant.² It is thus likely that among the first watches were some of spherical shape, the earliest dated example being a pomander watch, which formerly belonged to Philip Melanchthon.³ Where he acquired it is unknown and must remain a point of conjecture. Apart from this object only three cases of authentic pomander watches⁴ and a fragment are extant.⁵ Watches of spherical shape were produced in various cities in South Germany in the first half of the sixteenth century as well as in Blois.

About fifty examples of drum-shaped clocks are extant,⁶ and Henlein may have produced small versions of these as well. The height of construction depended on the duration of run (i.e., the length/number of turns of the fusee). It is noteworthy, that in a letter to Henlein written c. 1530 by Kaspar von Schöneich, Councillor to the Duke of Mecklenburg, a chain is mentioned in connection with the fusee for the first time.⁷ About 1550 the first neckwatches appeared.

Fig. 2. Henlein Fountain, designed by Max Meißner (b. 1859), bronze casting by Christoph Lenz, 1905 (Nuremberg, Hefnersplatz). Note the blacksmith's leg-vice attached to a tree trunk behind him.



3. Baltimore, The Walters Art Museum: 58.17. For a detailed survey of this extraordinarily important object see M. W. Gahtan and G. Thomas, 'Philip Melanchthon's Watch Dated 1530', *Antiquarian Horology*, 26 (2001/02), 377–388.

4. Two of these once belonged to the private collection of Otto Koch in Frankfurt a. M.; see E. v. Bassermann-Jordan, 'Ein Bisam-Apfel aus Peter Henleins Zeit', *Die Uhrmacher-Woche* 31, (1924), 301–2 (whereabouts unknown); id., 'Ein zweiter Bisam-Apfel aus Peter Henleins Zeit', *Die Uhrmacher-Woche*, 31 (1924), 627–8 (case without movement; private collection). The third case (formerly in the collection of George Daniels) is also private property, but currently on display in the Germanisches Nationalmuseum. The debate on the authenticity of a pomander watch allegedly of 1505 and ascribed to Henlein will not be addressed here (see below n. 16), and on the quarrels with its owner see T. Eser, *Die älteste Taschenuhr der Welt? Der Henlein-Uhrenstreit*, Nuremberg, Verlag des Germanischen Nationalmuseums, 2014, pp. 155–159.

5. A. Beck, 'Der Bisamapfel des Mathematisch-Physikalischen Salons', *Die Uhrmacher-Woche*, 49 (1942), 226–7 (case without movement, badly damaged in World War II).

6. For a tabular compilation see D. Matthes, 'Corpus der tragbaren deutschen Dosenuhren des 16. Jahrhunderts', in T. Eser, *Die älteste Taschenuhr der Welt?*, pp. 205–219; id., 'Die Gruppe der Henlein-Uhren', *Anzeiger des Germanischen Nationalmuseums*, 2018 (forthcoming).

7. P. Starsy, 'Ein Brief von Kaspar Schöneich an Peter Henlein', *Uhren und Schmuck*, 22 1985, 23–27, 54–56 (especially 25–26; for two more letters sent by Henlein to Schöneich found recently in the Landeshauptarchiv Schwerin see T. Eser, 'Die Henlein-Ausstellung im Germanischen Nationalmuseum. Rückblick, Ausblick, neue Funde', *Deutsche Gesellschaft für Chronometrie. Jahresschrift* 54 (2015), 31–33; an English version was published in *Antiquarian Horology*, 37 (2016), 199–212. Gruet of Geneva has been credited with the introduction of the fusee chain (e.g. W. I. Milham, *Time & Timekeepers Including the History, Construction, Care and Accuracy of Clocks and Watches*, New York, Macmillan, 1941, pp. 126, 224, 541), but no evidence has so far been found for this ascription (see O. Patrizzi, *Dictionnaire des horlogers genevois. La 'fabrique' et les arts annexes du XVIe siècle à nos jours*, Genève, Antiquorum, 1998, pp. 217–8).

The application of the stackfreed, an alternative construction for the fusee, allowed somewhat flatter movements, the oldest dated example being a neckwatch of 1548 signed CW (Caspar Werner, Nuremberg?). But the construction itself is older; it has been applied in the alarum of a table clock of 1533, and there are drawings of it by Leonardo extant.

In Germany a few cities took the lead in establishing the manufacture of clocks and instruments. In the course of the fifteenth century Nuremberg became a centre of clock- and instrument-making, as well as of the fine arts, mathematical, astronomical and geographical studies. Located at the intersection of major trade routes Nuremberg was one of Germany's largest cities with an extensive trade and was a thriving hub of elaborate craftsmanship. Raw materials (copper, zinc, tin, brass and bronze) were readily available, and Nuremberg's metal craftsmen and instrument makers were renowned for the quality of their work, from which also mathematicians, astronomers and cartographers profited. In 1471 Johannes Regiomontanus settled down in Nuremberg, established an observatory and a printing shop, from which numerous mathematical and astronomical works emerged. His pupil and patron Bernhard Walther is the first known person to have employed a clock for astronomical observations.⁸ With its diversified and highly developed craftsmanship Nuremberg seems indeed to be a likely location for the creation of the first portable timekeeper. Moreover, the necessary tools and equipment required for the manufacture of small-sized mechanisms (small files, drills &c.) could be acquired here easily.⁹ Little is still known about these 'external' factors however, and it is not unlikely that such favourable conditions may be encountered at other places also (e.g. Augsburg or Milan).

The beginnings of Henlein's apotheosis date back to the early nineteenth century, when Johann Ferdinand Roth (1748–1814), a Nuremberg theologian and local historian, published a paper on the so-called 'Nuremberg

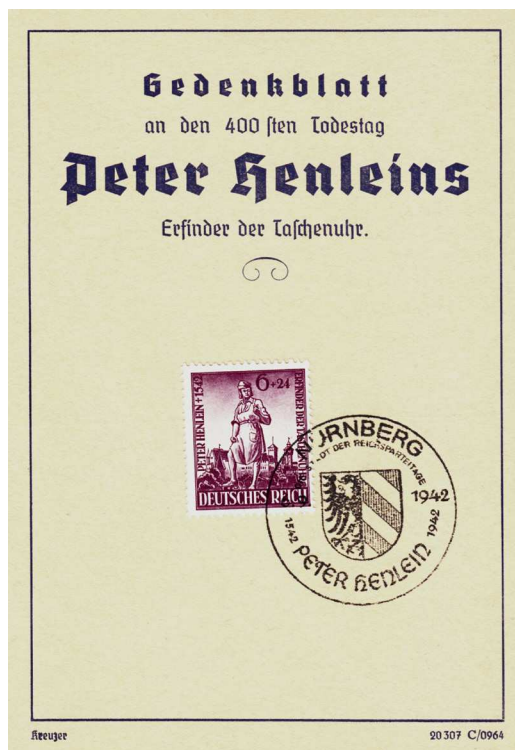


Fig. 3. Commemorative stamp sheet issued by the Reichspost on the 400th anniversary of Henlein's death, 1942 (in possession of the author).

Eggs' (a linguistic misinterpretation of 'Uhrlein'/Aeurlein', denoting a small timekeeper). In 1839 a short novel by Carl Spindler followed (*Der Nürnberger Sophokles*), which provided a model for later popular biographies. Towards the end of the nineteenth century the German watchmaker association (*Zentralverband der deutschen Uhrmacher*) and the engineer Franz Reuleaux (1829–1905) argued vigorously for the erection of a monument for Henlein, already labelled as 'the founding father of Germany's watch industry'. But an original watch made by Henlein had still not been found and was badly needed. This object appeared just in time – in 1897 the Germanisches Nationalmuseum acquired a drum-shaped watch, albeit of rather dubious provenance. The authenticity of this timepiece, signed *Petrus Hele me f. [ecit] Norimb[erga] 1510*, was subject

8. D. deB. Beaver, 'Bernard Walther. Innovator in Astronomical Observation', *Journal for the History of Astronomy*, 1 (1970), 39–43.

9. Nuremberg has been described as 'the ancestral city of filecutters' (*Ahnenstadt der Feilenhauer*), see O. Dick, *Die Feile und ihre Entwicklungsgeschichte*, Berlin, Julius Springer, 1925, pp. 23–31.

to serious doubts from the very beginning. It was soon discovered that the signature's letters had been copied exactly according to the typography of the edition of Pomponius Mela. Nevertheless the object was considered as the very first pocket watch of the world for a long time. It was incorporated in the fictional statue unveiled in Nuremberg in 1905, which portrays Henlein as a young man looking pensively at his creation. On this occasion an exhibition of historical timekeepers was presented in Nuremberg and a commemorative medal issued by the German Watchmaker's Association (*Deutscher Uhrmacherbund*).

The Henlein worship, amply underscored with pathetic and nationalistic claims, culminated in the infamous, entirely unhistorical hodgepodge of Veit Harlan's film *The Immortal Heart* (*Das unsterbliche Herz*) in 1939. The largely still intact old city of Nuremberg provided a splendid setting, and with crowds of inhabitants dressed in historical costumes dramatic mass scenes were staged. To commemorate the 400th anniversary of Henlein's death a special stamp was issued in 1942 (Fig. 3) Three years later the war was over, and Nuremberg lay in ruins.

Now the pendulum of interpretation swung in a different direction. In 1952 Enrico Morpurgo (Fig. 4) published a paper on Henlein ('L'orologio da petto prima del Henlein') in *La Clessidra*, followed by a booklet on the origin of the watch two years later.¹⁰ (Fig. 5) In his endeavour – without doubt spurred by the aggressive German claims of the thirties and very likely also by his own biography – Morpurgo took up the cudgels for Italy.

He was born in Trieste in 1894 and died in Amsterdam in 1972. At the end of the nineteenth century Trieste was a major harbour on the Dalmatian coast then belonging the Austro-Hungarian Monarchy and a city with a distinguished cosmopolitan cultural atmosphere. Morpurgo was born into an important and ancient Jewish family, whose name was derived from the city of Maribor in Slovenia, in former times called Marpurĝ. Already at the beginning of the



Fig. 4. Portrait of Enrico Morpurgo, undated photograph, 6.3 x 6.1 cm (© Universiteit van Amsterdam, Universiteitsmuseum, inv. nr. 131.525).

sixteenth century the family had members in Trieste and settled there permanently after the expulsion of the Jews from Vienna in 1560. Since 1721 they possessed a privileged status as *Hofjuden* and were active in trade, shipping, banking, as well as in the first insurance companies. Morpurgo attended Vienna University from 1913 to 1918, and in 1919 he received a PhD in Italian Literature in Padua for a study of the arias of the castrato Metastasio (*Le ariette di Pietro Metastasio*). Five years later he moved to the Netherlands, the native land of his Dutch wife. From 1935 to 1947 Morpurgo was a private lecturer (*Privatdozent*) in Italian at the University of Amsterdam and survived the Nazi occupation of the Netherlands although he was dismissed from the University.

From 1947 to 1953 he was a lecturer at the University of Groningen and afterwards he returned to Amsterdam, first as lecturer, then (in 1959) as an associate professor. In 1961 he was appointed as ordinary Professor of Italian. Morpurgo was a major contributor to the Italian horological magazine *La Clessidra* and published more than a hundred articles. He wrote a dictionary of Italian clockmakers,¹¹ as

10. *L'origine dell'orologio tascabile*, Rome, Edizioni 'La Clessidra', 1954. A later edition with the title *L'orologeria italiana dalle origini al Quattrocento* was published in Rome in 1986.

11. *Dizionario degli Orologiaisti Italiani (1300–1880)*, Rome, Edizioni 'La Clessidra', 1950.

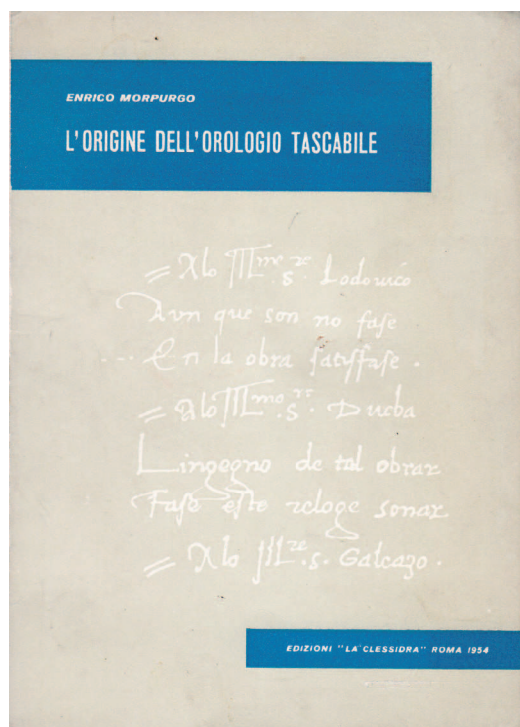


Fig. 5. Cover of Morpurgo's booklet *L'origine dell'orologio tascabile* (Rome 1954).

well as the equivalent text for Dutch makers.¹² Morpurgo looked eagerly for original sources in Italian archives to prove that portable timekeepers existed before the time of Peter Henlein. His interpretations were not always beyond doubt, and in his zeal to bear the palm for Italy he sometimes overstated his case. But he found that at the court of Ludovico Sforza small, portable clocks (*horologi piccoli e portativi*) and chiming timekeepers attached to ball costumes were mentioned around 1490.¹³ The German claims for Henlein's priority were also contested in Switzerland. In 1956 Léopold Defossez, director of the watchmaking school in Le Locle, published a paper in which he declared the historical impossibility of finding the inventor of the

pocket watch – be it German or Italian.¹⁴ The ensuing debates, however, were confined to German professional journals. Given the older assumptions, claims for alleged 'historical truth' and extremely, sometimes still unreflecting jingoistic interpretations, the issue was somewhat delicate in post-war Germany. In public awareness, Henlein's fame as the inventor of the pocket-watch remained firmly established.

With the consent of the Germanisches Nationalmuseum, a replica of the 'original drum-shaped watch by Peter Henlein' – with a Junghans movement! – was created for the 500th anniversary of Albrecht Dürer's birth by Karl Diehl KG of Nuremberg (then the majority shareholder of Junghans in Schramberg), and even the notorious inscription was faithfully copied. Quarrels broke out again when the German Watch and Clock Fair (INHORGENTA) celebrated the 500th anniversary of Henlein's birth in 1979. A second series of the 'Henlein watch' – now with a Junghans quartz movement – was produced. But the alleged year, 1479/80, was questioned and debated publicly and the Germanisches Nationalmuseum intervened. German horologists were mortally offended, and Jürgen Abeler (1993–2010), a well-known clock- and watchmaker and owner of a large collection of timekeepers in Wuppertal, published a pamphlet in 1980.¹⁵

In 2006 Ludwig Engelhardt, a watchmaker, teacher and local historian of Nuremberg, argued against all those who tried to deprive Henlein of his alleged glory as the one and sole inventor of a portable timekeeper. His main target was Morpurgo, and on his website (which is no longer accessible) Engelhardt raised profound accusations against him:

One realizes very quickly that Morpurgo does not want to create a neutral, serious, scientifically based research result – as an

12. *Nederlandse klokken- en horlogemakers vanaf 1300*, Amsterdam, Scheltema & Holkema, 1970.

13. E. Morpurgo, *L'origine dell'orologio tascabile*, pp. 38–40; see also G. Dohrn-van Rossum, *History of the Hour. Clocks and Modern Temporal Orders*, tr. Thomas Dunlap. Chicago/London, The University of Chicago Press, 1996, p. 121. Already in 1927 Marius Fallet had mentioned Italian letters dated 1466 and 1486, which probably referred to portable timekeepers, 'L'horloge portative à ressort a-t-elle existé avant Peter Henlein?', *Journal Suisse d'horlogerie*, 52 (1927), 245–6.

14. L. Defossez, 'A propos de l'origine de la montre', *Journal Suisse d'horlogerie*, 81 (1956), 175–180.

15. J. Abeler, *In Sachen Peter Henlein*, Wuppertal, Selbstverlag, 1980.

Italian he is keen on ascribing the first portable watch to an unknown fellow countryman, and apparently he is ready to use any means, even personal defamation, but this has nothing to do with scientific work.¹⁶

While his statement ‘All known reference works still mention Peter Henlein as the inventor of the pocket watch! There is no reason to change this!’¹⁷ can hardly be considered a convincing argument, Engelhardt was right in pointing out the weaknesses of some of Morpurgo’s assumptions and interpretations.

The late Giuseppe Brusa has aptly pointed out: ‘The controversy concerning the invention of watches always owed more to national pride than to evidence.’¹⁸ Too often wishful thinking and opinions based on dubious assumptions, even hearsay knowledge

were entangled when discussing Henlein’s life and achievements. Without doubt Henlein was an important historical figure and had a significant share in the development of portable timekeepers, but branding him as the sole and only inventor is an inappropriate exaggeration. More likely the miniaturisation of clock mechanisms seems to be a silent transition occurring at different places more or less contemporaneously, and therefore all efforts to discover ‘the oldest watch’ are futile. Sometimes very little is left after a critical assessment of the original sources, and, like in theology, such demythologization incites those who pretend to know ‘the truth’ to ferocious discussions. So it is rather unlikely that any attempt to a sober and balanced approach – the recent exhibition of the Germanisches Nationalmuseum in 2014–15 being a particular fine example¹⁹ – will put an end to disputes once and for all.

16. ‘[...] man [merkt] sehr schnell, dass Morpurgo kein neutrales, seriöses, wissenschaftlich fundiertes Forschungsergebnis erstellen will – ihm geht es als Italiener darum, einem unbekannten Landsmann die erste tragbare Uhr zuzuschreiben und da ist ihm anscheinend jedes Mittel recht, auch persönliche Diffamierung – aber dies hat mit einer wissenschaftlichen Arbeit nichts zu tun’. This is a quotation from Engelhardt’s website, accessed in 2014. In a privately published booklet [*Peter Henlein Taschenuhr Anno 1505*, Nuremberg n.d.] kindly provided to the author, similar arguments are found.

17. ‘Alle bekannten Nachschlagewerke kennen aber immer noch Peter Henlein als Erfinder der Taschenuhr! Es gibt keinen Grund dies zu ändern!’

18. G. Brusa, ‘Early Mechanical Horology in Italy’, *Antiquarian Horology*, 18 (1989/90), 510; see also id., ‘I primi orologi da persona in Italia. Nuovi indizi e nuove eccellenti testimonianze’, *La Voce di Hora. Pubblicazione dell’Associazione Italiana Cultori di Orologeria Antica*, nr. 3, (1997), 3–20.

19. T. Eser, *Die älteste Taschenuhr der Welt?* and ‘Die Henlein-Ausstellung im Germanischen Nationalmuseum’ (see n. 7), 23–34.